Jane Doe

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Springfield, ST | US Citizen | Willing to relocate

Education

B.S. Electrical Engineering, GPA 3.62/4.00 Expected Spring 2027
Example State University
Relevant coursework: Power Electronics, Embedded Systems, Digital Design, Signals and Systems, Control Systems
Activities: IEEE student branch, Tau Beta Pi

Projects

Synchronous Buck Converter for a CubeSat Power Board Fall 2025–Spring 2026
KiCad, LTspice, STM32, C

- Reached 92% peak efficiency at a 2 A load, measured on an electronic load, by replacing the catch diode with a synchronous MOSFET stage and tuning dead time with an STM32 timer
- Cut output ripple from 80 mV to 25 mV peak to peak, measured on an oscilloscope, by redesigning the 4-layer PCB in KiCad with a solid ground plane and a tighter switching loop
- Wrote STM32 firmware in C that shuts the rail down within 50 µs of an overcurrent fault, verified over 20 deliberate short-circuit tests

FPGA Audio Equalizer Spring 2025
Verilog, Vivado, MATLAB, Basys 3

- Processed 48 kHz stereo audio at under 2 ms latency by pipelining five 64-tap FIR filters in Verilog
- Matched the MATLAB reference response within 0.5 dB from 20 Hz to 20 kHz by testing fixed-point coefficients in Vivado simulation

Bluetooth Battery Monitor Fall 2024
ESP32, C++, Python, INA219

- Tracked four Li-ion cells to within 10 mV each with an ESP32 monitor that streams readings over Bluetooth to a Python dashboard
- Measured pack current within 1% of a bench multimeter by calibrating an INA219 sensor across a 0.1 A to 3 A load sweep
- Extended field-test runtime by 18% by adding a low-voltage cutoff in C++ that sheds non-critical loads at 3.3 V per cell

Experience

Power Subsystem Lead, Example State CubeSat Team, Springfield, ST Sep 2024–Present

- Cut power board mass by 40 g by replacing three linear regulators with one buck converter board
- Found two battery protection faults before launch review by writing a Python script that runs overnight charge and discharge cycles and flags cell drift
- Led weekly design reviews for a four-person power team and kept board revisions on schedule for two launch-readiness milestones

Sales Associate, Example Hardware Store May 2023–Aug 2024

Skills

Software: KiCad, LTspice, MATLAB, Vivado, Python, C, C++, Verilog
Hardware/Lab: Oscilloscope, Electronic load, STM32, ESP32, FPGA, Soldering
Certifications: IPC J-STD-001 Soldering